

Certificate Template Management & Generation System

1. Project Overview

Objective

Develop a web-based certificate management system that enables administrators to:

- Upload certificate templates.
 - Store templates permanently.
 - Automatically detect placeholder positions using AI.
 - Fill certificate details such as recipient name and dates.
 - Preview generated certificates.
 - Download certificates in PDF or PNG format.
 - Share certificates through WhatsApp using the device sharing mechanism.
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2. Functional Requirements

Module 1: Template Management

Features

- Upload certificate templates (PNG/JPG).
- Store templates permanently.
- Display uploaded templates in gallery view.
- View template details.
- Delete templates.
- Update template metadata.

Accepted Formats

- PNG
- JPG
- JPEG

Validation

- Maximum file size: 10 MB
 - Minimum resolution: 1000 × 700
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Module 2: AI-Assisted Placeholder Detection

Objective

Automatically identify certificate placeholders.

Supported Placeholders

- Recipient Name
- From Date
- To Date

Workflow

1. Admin uploads certificate template.
2. Template image sent to Gemini Vision.
3. Gemini returns suggested coordinates.
4. OpenCV validates detected regions.
5. Coordinates displayed on template.
6. Admin confirms or adjusts positions.
7. Final coordinates stored.

Output

Example:

```
{ "recipient_name": { "x": 540, "y": 310 }, "from_date": { "x": 420, "y": 560 }, "to_date": { "x": 720, "y": 560 } }
```

Module 3: Template Configuration

Features

Admin can:

- Move placeholder positions
- Resize text regions
- Set font size
- Set font family
- Set font color
- Save configuration

Configuration Saved Per Template

Recipient Name

- X Coordinate

- Y Coordinate
- Font Size
- Font Family
- Color

From Date

- X Coordinate
- Y Coordinate
- Font Size
- Font Family
- Color

To Date

- X Coordinate
- Y Coordinate
- Font Size
- Font Family
- Color

Module 4: Certificate Generation

Input Fields

Recipient Name From Date To Date

Process

1. Load template image.
2. Retrieve placeholder coordinates.
3. Render text onto template.
4. Generate preview.
5. Confirm generation.

Output

- PNG Certificate
- PDF Certificate

Module 5: Certificate Preview

Features

- Real-time preview

- Zoom support
 - Full-screen preview
 - Validate text placement
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Module 6: Download

Supported Formats

PNG PDF

Actions

Download PNG Download PDF

Module 7: WhatsApp Sharing

Mobile Devices

Use Web Share API.

Workflow:

Generate Certificate → Share → WhatsApp → Select Recipient → Send

Desktop

Download certificate first. User manually uploads to WhatsApp Web.

3. Non-Functional Requirements

Performance

Template Upload

< 5 seconds

Certificate Generation

< 2 seconds

Preview Rendering

< 1 second

Security

- JWT Authentication
 - Password Hashing
 - File Validation
 - Input Validation
-

Scalability

Current:

- Local File Storage

Future:

- AWS S3
 - CDN
-

4. Technology Stack

Frontend

React.js

Purpose: User Interface

Libraries:

- React Router
 - Axios
 - Material UI
 - React Hook Form
 - React Konva
-

Backend

FastAPI

Purpose: REST API

Server:

Uvicorn

Database

MySQL

ORM:

SQLAlchemy

Migration:

Alembic

AI Layer

Gemini 2.5 Flash Vision

Purpose:

- Detect placeholders
 - Suggest coordinates
-

Computer Vision

OpenCV

Purpose:

- Underline detection
 - Bounding box detection
 - Coordinate refinement
-

Image Processing

Pillow (PIL)

Purpose:

- Load images
 - Render text
 - Generate PNG
-

PDF Generation

ReportLab

Purpose:

- Convert certificates into PDF
-

Storage

Development

Local Storage

uploads/templates/ uploads/generated/

Production

AWS S3 (Optional)

Deployment

Frontend

Vercel

Backend

Render / VPS / EC2

Database

MySQL

5. Database Design

users

id username email password_hash role created_at

templates

id template_name image_path thumbnail_path created_at

placeholder_config

id template_id field_name x_coordinate y_coordinate font_size font_family font_color

generated_certificates

id template_id recipient_name from_date to_date certificate_path created_at

6. REST API Specification

AUTHENTICATION

POST /api/auth/login

Request

```
{ "email": "admin@example.com", "password": "password123" }
```

Response

```
{ "token": "jwt_token" }
```

TEMPLATE MANAGEMENT

POST /api/templates/upload

Request

Multipart Form Data

template_image

Response

```
{ "template_id": 1, "message": "Template uploaded successfully" }
```

GET /api/templates

Response

```
[ { "id": 1, "name": "Certificate Template", "image_url": "/uploads/templates/template1.png" } ]
```

GET /api/templates/{id}

Response

```
{ "id": 1, "name": "Certificate Template", "image_url": "/uploads/templates/template1.png" }
```

DELETE /api/templates/{id}

Response

```
{ "message": "Template deleted successfully" }
```

AI TEMPLATE ANALYSIS

POST /api/templates/{id}/analyze

Response

```
{ "recipient_name": { "x": 540, "y": 310 }, "from_date": { "x": 420, "y": 560 }, "to_date": { "x": 720, "y": 560 } }
```

PLACEHOLDER CONFIGURATION

POST /api/templates/{id}/placeholders

Request

```
{ "recipient_name": { "x": 540, "y": 310, "font_size": 42 }, "from_date": { "x": 420, "y": 560, "font_size": 18 } }
```

Response

```
{ "message": "Configuration saved" }
```

CERTIFICATE PREVIEW

POST /api/certificates/preview

Request

```
{ "template_id": 1, "recipient_name": "John Doe", "from_date": "2026-01-01", "to_date": "2026-06-30" }
```

Response

```
{ "preview_url": "/generated/preview_123.png" }
```

CERTIFICATE GENERATION

POST /api/certificates/generate

Request

```
{ "template_id": 1, "recipient_name": "John Doe", "from_date": "2026-01-01", "to_date": "2026-06-30",  
  "format": "pdf" }
```

Response

```
{ "certificate_id": 101, "download_url": "/generated/certificate101.pdf" }
```

DOWNLOAD

GET /api/certificates/{id}/download

Response

Binary File

SHARE

GET /api/certificates/{id}/share

Response

```
{ "file_url": "/generated/certificate101.pdf" }
```